

Utilization Patterns and Perceived Experience in Campus Recreation Centers

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SBE 498 Senior Capstone
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Spring 2026

Abstract

Campus recreation centers are often evaluated through broad indicators such as capacity and facility performance, yet these measures do not fully explain how users experience the space during actual use. Existing research has given less attention to how equipment demand, spatial organization, and user perception interact within high-traffic fitness environments. This study addressed that gap by examining how machine utilization patterns relate to perceived user experience across two University of Arizona recreation centers. A mixed-method observational design was used, combining structured observations of cardiovascular equipment usage with intercept surveys administered during midday and evening time blocks. Utilization was categorized into low, moderate, and high conditions, while survey responses captured perceptions of crowding, comfort, movement, and equipment access. Results showed that overall machine utilization remained low across both facilities and time blocks, yet users consistently reported moderate levels of crowding and experienced measurable limitations in equipment access. Machine-level analysis indicated that demand concentrated around specific equipment types, particularly treadmills and stair-based machines, while other resources remained underutilized. Behavioral responses further showed that users often waited within preferred zones rather than relocating elsewhere in the facility. These findings suggest that user experience is shaped more by localized demand patterns, equipment distribution, and circulation than by the total system capacity alone. The study underlines the importance of integrating behavioral and perceptual measures into the design and evaluation of recreation centers, especially those seeking to balance environmental performance with user-centered outcomes.

Keywords: campus recreation centers, user experience, equipment utilization, spatial organization, indoor environmental quality, sustainable design, LEED

Introduction

The Built Environment as a Behavioral Influence in Fitness Spaces

Fitness and recreation centers are structured to promote health, performance, and sustained participation. Yet, the environments in which exercise happens shape how activity is experienced. Thermal comfort, ventilation effectiveness, lighting distribution, spatial organization, and equipment placement influence perceived effort, concentration, and enjoyment. Research in environmental psychology and building science demonstrates that indoor environmental conditions influence stress, cognitive clarity, and overall comfort perception (Thach et al., 2020; Kim et al., 2022). Environmental conditions therefore act as behavioral variables, influencing where users choose to engage and exercise, how long they do so, and the perceived quality of their workouts.

Recreation centers present a distinct context for examining this influence. Exercise elevates metabolic heat production, respiratory demand, and sensitivity to airflow and temperature variation increasing sensitivity to environmental differences (Avci et al., 2024). Users also move between different zones within the facility, each with varying occupancy and environmental conditions. As a result, environmental quality is not experienced uniformly across the building. Prior research suggests that environmental conditions can contribute to clustering behavior, where users move toward more favorable areas that feel more comfortable (Shin et al., 2013; Cao et al., 2025).

In this study, “sustainable” refers to the ability of a space to benefit user’s comfort and long-term engagement. Leadership in Energy and Environmental Design (LEED) prioritizes Indoor Environmental Quality through credits addressing ventilation, daylight access, thermal regulation, and low-emission materials. Empirical evaluations of LEED-certified buildings frequently report improved satisfaction and perceived environmental quality (Fissore et al., 2023; Lawal et al., 2025). Post-occupancy research highlights that occupant satisfaction often depends on the interaction of multiple IEQ factors, emphasizing the importance of examining spatial distribution rather than only aggregate metrics (Huang et al., 2022). Institutions invest in LEED certification to signal environmental responsibility and long-term operational efficiency. The University of Arizona South Recreation Center (SouthREC), recognized as the first collegiate recreation center to achieve LEED Platinum certification, exemplifies this commitment. Prior research suggests that even within certified buildings, localized zones of discomfort can affect engagement, especially under high metabolic load or intense physical activity (Salonen et al., 2020).

Self-Determination Theory identifies autonomy, competence, and relatedness as key factors that support sustained physical activity (Teixeira et al., 2012). Individuals who maintain long-term exercise routines often report higher enjoyment and a stronger sense of competence in their workouts compared to those who disengage (Geller et al., 2018). Environmental conditions can influence these factors by affecting how comfortable, controllable, and supportive a workout space feels. For example, a space with proper airflow and a comfortable temperature may make exercise feel more manageable, which then reinforces a sense of competence. In contrast, uncomfortable or crowded spaces may increase perceived difficulty and reduce overall motivation. Research also shows that overly restrictive or uncomfortable

environments can reduce intrinsic motivation by shifting the user's attention away from enjoyment and toward discomfort (Vansteenkiste et al., 2007). Supportive environments, on the other hand, can enhance motivation and improve well-being (Gidney et al., 2024; Lolli et al., 2022).

Despite advancements in both sustainable building research and exercise psychology, few studies directly connect environmental performance metrics with observed behavioral patterns within high-intensity recreation environments. This study addresses that gap by examining how spatially distributed environmental conditions relate to perceived comfort, equipment usage patterns, and user experience across two University of Arizona campus recreation facilities. By integrating observational data with user perception, this research contributes to a more applied understanding of how sustainable design can influence behavior and engagement in fitness spaces.

Indoor Environmental Quality as a Distributed and Perceived Condition

Indoor Environmental Quality (IEQ) research increasingly emphasizes perceived comfort as a key outcome of building performance. Comfort does not arise from a single environmental variable, but from the interaction of thermal, visual, acoustic, and air quality conditions within specific spatial contexts (Fissore et al., 2023). These findings highlight the importance of evaluating environmental quality as spatially distributed rather than uniform across an entire structure. Psychological outcomes further reinforce this perspective, as higher satisfaction across IEQ domains corresponds with lower self-reported stress levels, even when buildings meet technical standards (Thach et al., 2020). Studies of university facilities demonstrate persistent dissatisfaction with thermal and air quality conditions despite regulatory compliance, indicating that environmental experience varies within zones of the same building (Kim et al., 2022). These spatial gradients are particularly relevant for recreation centers, where clustering and metabolic demand can fluctuate throughout the day. Airflow patterns and thermal variability can influence environmental favorability, affecting where users may choose to exercise (Salonen et al., 2020; Cao et al., 2025).

Exercise introduces physiological conditions that increase sensitivity to environmental variation. As body temperature and breathing increase during activity, users become more sensitive to airflow and temperature changes, meaning standardized predictors of comfort are not always able to reflect how conditions are being experienced (Avci et al., 2024). These findings show the importance of considering both activity intensity and user adaptation when evaluating IEQ. Studies also link thermal discomfort to reduced motivation and altered activity patterns, demonstrating the value of integrating behavioral data with environmental metrics (Woo et al., 2021; Shin et al., 2013).

Spatial Organization and Behavioral Concentration in Fitness Facilities

The built environment influences physical activity behavior through accessibility, layout, and environmental functionality. Within fitness facilities, users prioritize adequate ventilation, sufficient exercise space, and equipment reliability when evaluating quality (Yong Jang, 2024). However, existing studies often rely on generalized satisfaction surveys or macro-level facility assessments. Few investigations examine how daylight distribution, airflow patterns, or thermal variations correspond with

local patterns of equipment usage. Large recreation centers frequently contain multiple activity zones that differ in environmental exposure due to facade orientation, HVAC zoning, or glazing strategies associated with sustainable or LEED-informed design. These patterns are amplified in high-traffic environments, where environmental cues may interact with social clustering and equipment selection (Shin et al., 2013; Cao et al., 2015).

Observational findings from the SouthREC pilot study revealed concentrated use of treadmills and stair machines, alongside underutilized equipment in peripheral zones. Understanding whether high-demand equipment aligns with environmentally favorable zones could inform evidence-based facility planning. If comfort and motivation are influenced by environmental exposure, equipment placement may represent a key factor for optimizing facility engagement.

Motivational Factors within Environmental Context

Self-Determination Theory provides a robust framework for examining sustained exercise behavior. Intrinsic motivation, characterized by enjoyment and personal value, predicts long-term adherence more strongly than external motives (Teixeira et al., 2012). Longitudinal studies indicate that individuals who maintain consistent activity levels report higher perceived competence and enjoyment compared to those who disengage (Geller et al., 2018). Among university populations, intrinsic motivation remains positively associated with higher activity participation across genders (Alecu et al., 2025). Further experimental research has demonstrated that externally controlling contexts can reduce engagement by shifting focus away from intrinsic satisfaction and toward performance evaluation (Vansteenkiste et al., 2007). Environmental discomfort can similarly increase perceived strain or distraction, reducing engagement and enjoyment.

Environmental factors such as daylight, airflow, thermal regulation, and acoustic quality can reinforce or undermine these motivational mechanisms by influencing perceived comfort and autonomy (Fissore et al., 2023; Thach et al., 2020). Supportive environmental elements contribute to more self-determined motivational profiles, linking comfort directly to engagement and persistence (Gidney et al., 2024; Lan et al., 2023). In high-intensity settings, even minor improvements in environmental favorability, like enhanced ventilation or better daylighting, may improve perceived competence and willingness to continue exercising (Avci et al., 2024).

Comparisons between indoor and outdoor exercise environments further suggest that environmental qualities shape psychological responses, although longitudinal findings remain limited (Noseworthy et al., 2023). LEED-informed indoor spaces may replicate some of the restorative and motivational benefits of outdoor environments when daylight, thermal comfort, and airflow are thoughtfully distributed (Lawal et al., 2025; Wang et al., 2024).

Identifying the Gap

Collectively, the literature demonstrates that indoor environmental quality influences perceived comfort and stress, spatial organization affects facility satisfaction, and intrinsic motivation predicts

sustained exercise engagement. However, research connecting spatial environmental conditions with both observed behavioral patterns and motivational outcomes in university recreation centers remains limited, particularly for comparisons between LEED-certified and non-certified facilities.

The present study expands this preliminary work by comparing a LEED-certified recreation center with a non-certified campus facility. It examines how spatially distributed IEQ characteristics relate to perceived comfort, motivational orientation, and equipment usage. By combining environmental performance analysis with exercise psychology constructs, this study develops a more integrated understanding of sustainable design in fitness facilities and identifies conditions where environmental strategies can enhance both engagement and psychological satisfaction.

Methodology

Research Design and Question

This study examines how machine utilization patterns and perceived user experience vary across recreation centers and time-of-day conditions. The primary objective is to understand how measured system demand relates to how those same conditions are experienced by users, and what that relationship suggests about spatial organization and behavioral engagement with fitness environments.

A mixed-method observational design was used to capture both sides of this relationship. Structured observations of cardiovascular equipment provide a consistent measure of how the system is being used at specific points in time. Intercept surveys capture how those same conditions are interpreted by users in terms of crowding, comfort, movement, and equipment access. These two components are considered together so that observed usage can be directly compared with user experience under the same conditions. This study builds on prior pilot work conducted at a single facility and expands to include both NorthREC and SouthREC. Comparing two recreation centers introduces variations in layout, circulation, and equipment distribution, which allows patterns to be interpreted beyond a single, spatial context. The focus is shifted away from evaluating individual environmental features and instead looks at how the overall organization of the space shapes where users concentrate, how they move through the facility, and how they experience it during typical operating conditions.

Data Collection

Data were collected from February 28 through March 21 across both recreation centers, with observation centers conducted Monday through Saturday. Two time blocks were selected, 11:00-11:30 AM and 5:00-5:30 PM, based on prior experience working in both facilities where these periods consistently represent moderate and peak demand. Rather than assigning fixed times to each location, the schedule alternated daily so that both facilities were observed under similar conditions. During each session, cardiovascular machines were recorded by type. For each category, the total number of units and the number in use were documented at the time of observation. Utilization was calculated as the percentage of machines in use relative to those available. These values were then grouped into low,

moderate, and high utilization using thresholds of below 30 percent, between 30 and 60 percent, and above 60 percent, respectively. Grouping utilization in this way allows patterns to be compared more clearly across sessions, while also reflecting more meaningful differences between underuse, typical operations, and congestion within the facilities.

Survey data were collected immediately following facility use using a brief intercept approach. The first ten individuals exiting the facility during each session were approached and asked to respond. This method provided a consistent sample across sessions while keeping the process efficient and minimally disruptive. The survey focused on what users intended to do during their visit, whether they were able to access the equipment they planned to use, and how they chose to adjust their visit if they could not. Additional questions were asked regarding perceptions of crowding, comfort, and ease of movement, and to identify their primary reasons for choosing that facility. Responses were recorded directly by the researcher and kept anonymous. No identifying information was collected, and participation was voluntary.

Data Triangulation and Analysis Approach

The analysis is structured around comparing observed conditions with reported experience. Observational data captures how equipment is being used and how often different levels of utilization occur across times and locations. Survey responses provide further insight into how those same conditions are experienced by users. Rather than treating these as separate findings, the analysis focuses on how they relate to one another. Utilization rates are compared with perceived crowding to determine whether higher levels of use correspond to a stronger sense of congestion. Equipment access responses are evaluated alongside those same patterns to identify situations where users report difficulty even when the overall utilization remains low. These conditions are interpreted as localized constraints within the recreation centers rather than indicators of maximum capacity.

Movement and comfort responses are also considered in how they relate to the observed usage patterns to better comprehend how users are navigating the space. In particular, the analysis examines whether users relocate across the facility when conditions change or remain within specific areas and locally adjust their behaviors. This provides insight into how spatial organization influences user behavior beyond what can be observed and captured through utilization data alone.

Data Preparation

The dataset was organized in Google Sheets prior to analysis, with observational and survey data structured to allow for direct comparison across facility and time blocks. This included calculating utilization rates, categorizing observations, and organizing survey responses with consistent variables to summarize through pivot tables and frequency counts. The structuring of the dataset required the alignment of multiple layers of information including machine type, time block, facility, and survey responses. Observation data and survey responses were linked by session, allowing each time block to be

analyzed using both behavioral and perceptual data. This made sure that utilization patterns could be analyzed alongside the reported experience under the same conditions.

Results

Overall Utilization Patterns

Average machine utilization remained low across both recreation centers and time blocks. Across all observed conditions, utilization values showed only minor variation between NorthREC and SouthREC, as well as between the 11AM and 5PM sessions. While the 5PM time block did consistently produce slightly higher levels of use, the increase was modest and not indicative of a substantial change in overall demand. Both NorthREC and SouthREC also exhibited comparable utilization levels, with neither location approaching thresholds that would suggest widespread congestion. This consistency across centers suggests that the observed differences in user experience are not driven by large differences in total equipment demand between the two facilities.

At no point did system-wide usage approach levels where the majority of equipment was in use simultaneously. Instead, the data reflects conditions in which significant portions of available equipment remained unused across most of the sessions. These patterns indicate that, at a broad level, cardiovascular capacity was not fully utilized during the observation periods. The system as a whole did not operate under conditions that would suggest overall congestion, even during peak hours.

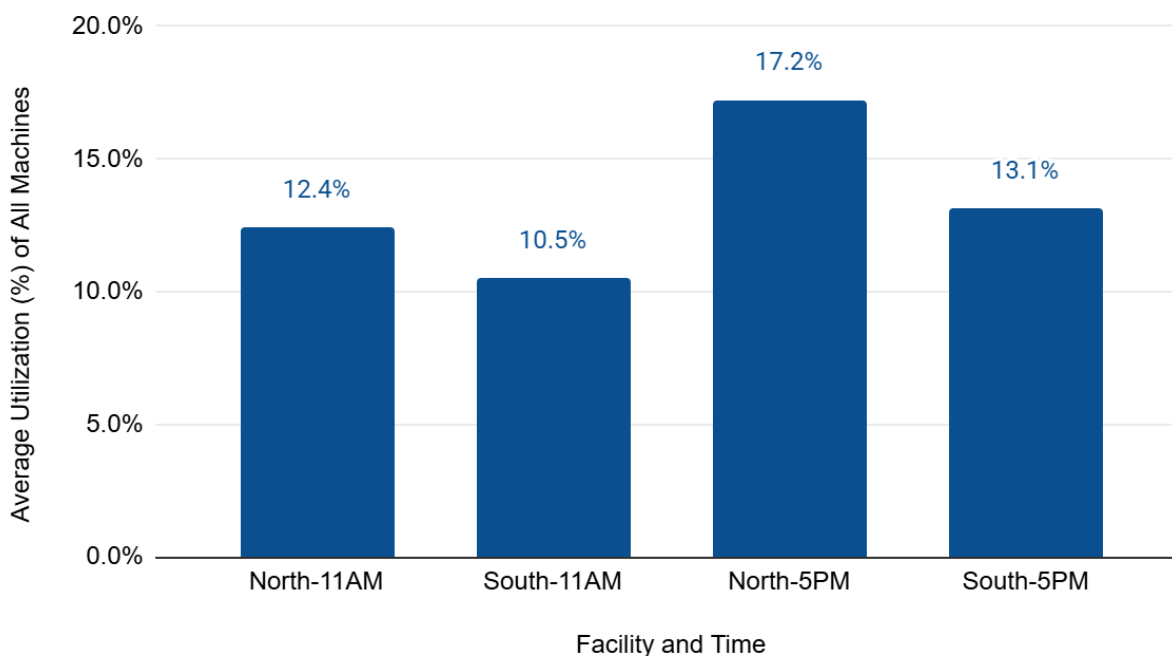


Figure 1A. Average Rate of Machine Utilization

Distribution of Utilization Conditions

When utilization is examined by category rather than the average values, a consistent pattern emerges across both facilities and temporal conditions. The majority of observations were classified as low utilization, with moderate conditions appearing less frequently and high utilization occurring only in isolated instances. This distribution was held across both NorthREC and SouthREC, as well as across the 11AM and 5PM observation times. The facility itself did not fluctuate between low and high demand in a way that would indicate unstable or peak-driven usage. Conditions remained within the lower end of the utilization range, maintaining stability with occasional moderate and short-lived high utilization.

This pattern is illustrated in Figure 1B, where low utilization consistently represents the largest portion of observations across all conditions. Moderate utilization appears occasionally for each temporal block, while high utilization shows up only a few times, except for at NorthREC during 11AM observations. The relative proportions are similar between the facilities, further indicating that the observed conditions reflect a stable pattern within the system. This consistency is important to distinguish between isolated moments of congestion and broader facility conditions. While periods of higher demand were observed, they were not sustained and did not define the normal patterns of use. The overall distribution indicates that higher-demand conditions were the exception, with most sessions operating under low utilization.

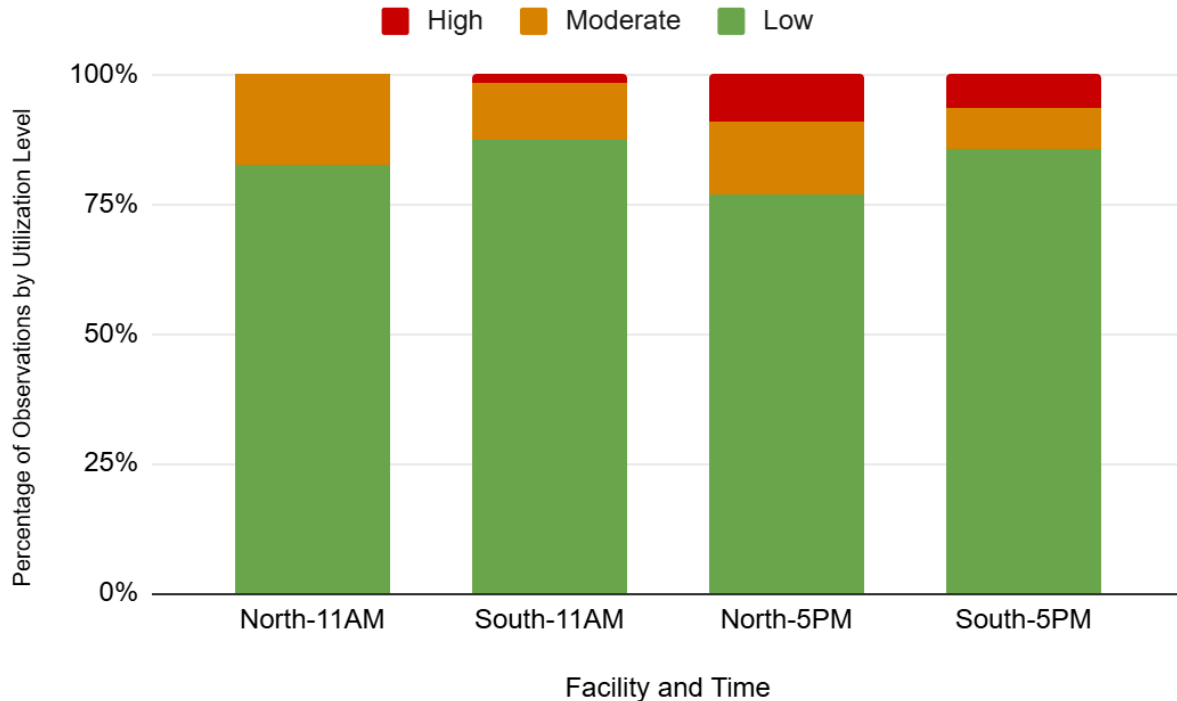


Figure 1B. Distribution of Utilization Conditions

Machine-Level Utilization

While overall utilization remained low at the system level, machine-level patterns show that demand was not evenly distributed across equipment types. As shown in Table 1, usage is concentrated among a small number of machines, while others remain largely or completely unused across both facilities and times. Treadmills and stairmasters exhibited the highest levels of use. For example, treadmill utilization ranged from 33.3% to 46.6% across conditions, reaching its highest usage during the SouthREC 5PM sessions. Stairmaster usage followed a similar pattern, with values ranging from 28.7% to 44.1%, also peaking during the 5PM block.

In contrast, several machine types remained consistently underutilized. Ellipticals showed relatively low usage, ranging from 3.0% to 13.3%, while stationary bikes remained below 10% across all conditions. Other equipment categories, including Arc Trainers and hand pedals, recorded no usage during the observed periods. These differences in utilization indicate that users are not evenly distributed across available equipment and are concentrated around specific machines.

The uneven distribution across machine types helps to explain how localized congestion may occur even when the overall utilization within the facility is low. Higher demand machines experience repeated uses within the same time periods, which limits availability despite facility capacity. As a result, users may experience difficulty in accessing specific equipment even if the rest of the facility is exhibiting underutilization. These patterns suggest that perceptions of crowding and access limitations are more closely tied to machine-level demand instead of overall facility conditions.

Table 1. Average Machine Utilization by Type, Facility, and Time Block

Machine Type	North (11AM)	South (11AM)	North (5PM)	South (5PM)
Arc Trainer	0.0%	0.0%	0.0%	0.0%
Elliptical	10.5%	4.0%	13.3%	3.0%
Hand Pedals	0.0%	0.0%	0.0%	0.0%
Miscellaneous	0.0%	0.0%	0.0%	0.0%
Stairmaster	36.1%	28.7%	44.1%	32.6%
Stationary Bike	6.6%	7.4%	5.9%	9.6%
Treadmill	33.7%	33.3%	39.3%	46.6%

Perceived Experience

Survey responses indicate that users did not consistently experience the facilities as underutilized, despite low machine usage. Reported crowding generally fell in a moderate range, while comfort and ease of movement were considered positive across conditions. These results suggest that perceived experience does not directly align with utilization levels, and that users may be responding more to localized conditions within the facility.

As shown in Figure 2, average crowding ratings ranged from 2.4 to 3.0 across all conditions, with the highest value occurring at NorthREC during the 5PM time block. Comfort ratings were relatively stable, ranging between 3.2 to 3.5, while movement was similar with ratings from 3.1 to 3.5. These values suggest that even as crowding began to approach moderate levels, users reported generally favorable conditions for comfort and movement. This separation between perceptions further supports the idea that different aspects of user experience are influenced by different spatial conditions.

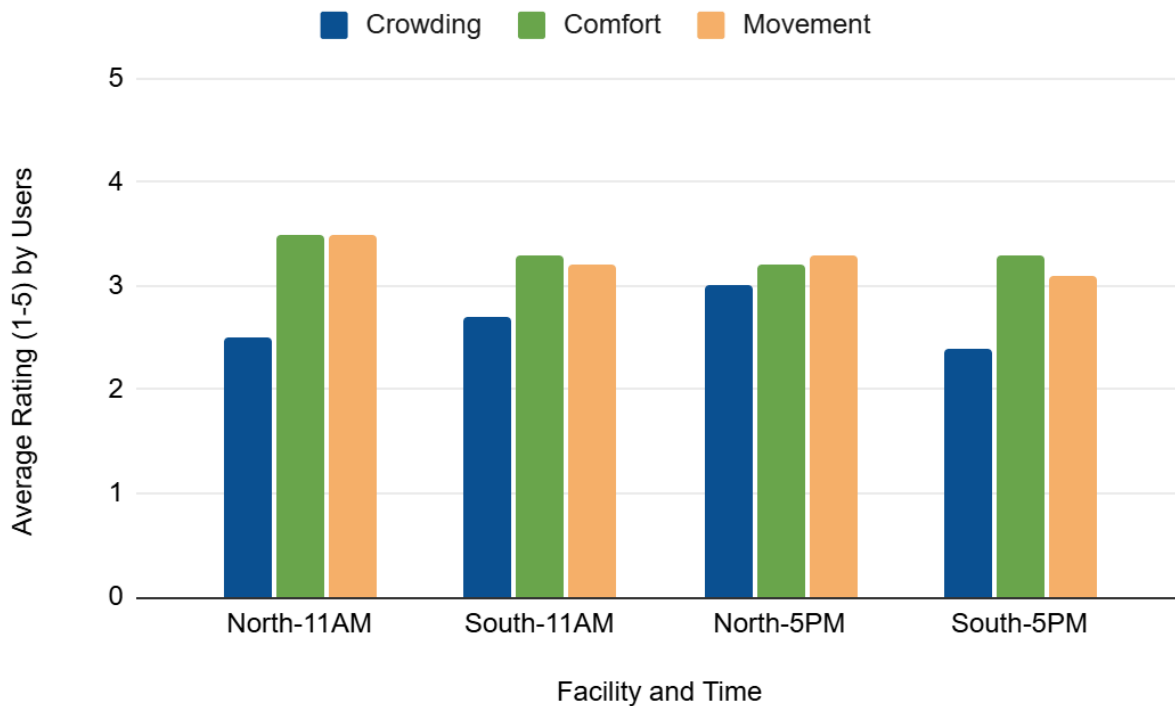


Figure 2. Perceived Crowding, Comfort, and Movement by Facility and Time Block

Equipment Access and Behavioral Response

Most responses reported being able to access the equipment they intended to use, though a notable amount indicated that they were unable to do so. As shown in Figure 3, access rates were highest during the 11AM time block, with 86.36% of users at NorthREC and 84.00% at SouthREC reporting successful access. At 5PM, access decreased to only 71.43% of users at NorthREC and 73.68% at SouthREC reporting access, corresponding to increases in reported inaccessibility of up to 28.57% and 35.71% respectively. The presence of access issues alongside low demand suggests that constraints are concentrated among specific machines and areas.

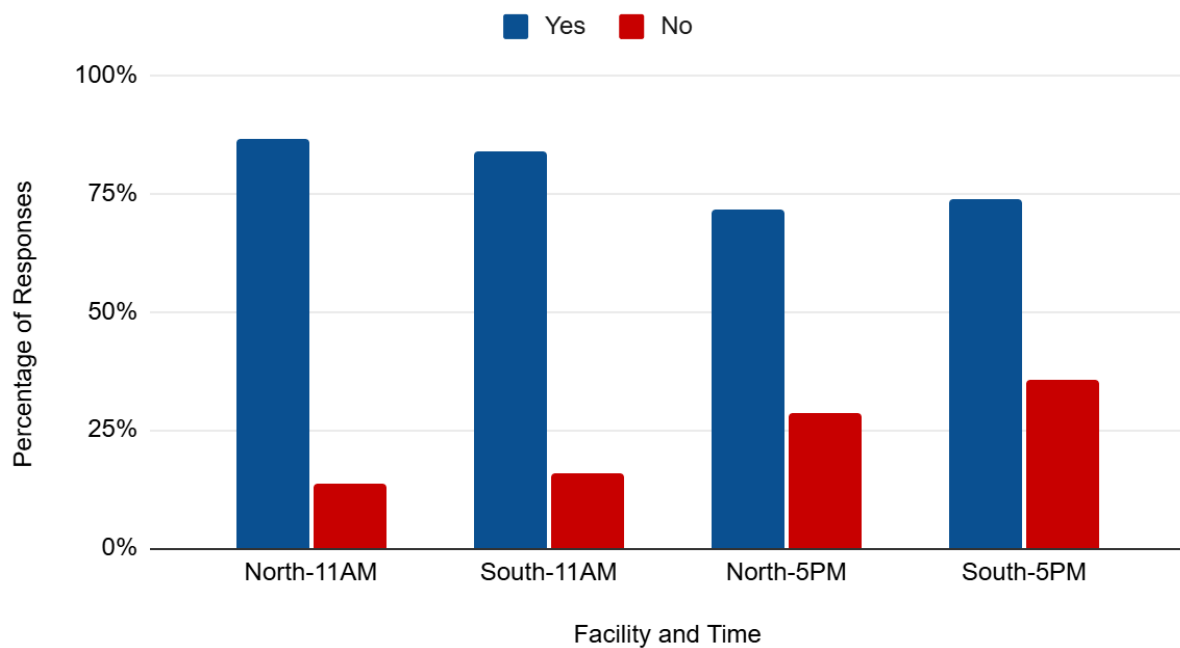


Figure 3. Equipment Access by Facility and Time Block

When equipment was unavailable, users most commonly reported waiting rather than relocating to another area of the recreation center. Figure 4 shows that 50.0% of respondents who experienced equipment inaccessibility chose to wait, while other adjustments tended to occur less frequently. Shortening workouts accounted for 16.7% of responses, while switching machines and skipping planned activities each accounted for 12.5%. Movement across the facility itself showed to be minimal, with only 4.2% of users reporting that they switched areas. These patterns indicate that responses to limited access are localized, with users adapting within the same area rather than dispersing across the facility.

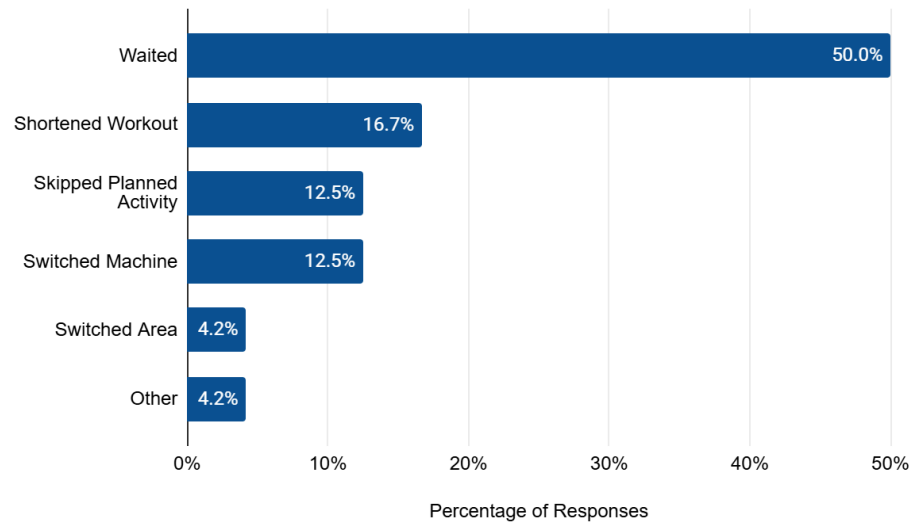


Figure 4. Behavioral Adjustments Following Equipment Inaccessibility

Facility Selection Factors

Survey responses indicate that users primarily selected a recreation center based on convenience and schedule. As shown in Figure 5, convenience was selected most frequently among the factors (63 responses), followed closely by schedule (56 responses). Secondary factors like habit (23) and social reasons (22) were present but far less dominant. In contrast, factors related to the built environment were selected much less frequently with less crowded receiving 12 responses, layout 6, and equipment 4. These results are suggestive of spatial conditions not being the primary drivers of initial facility selection. Instead, users appear to prioritize accessibility and routine over design-related considerations.

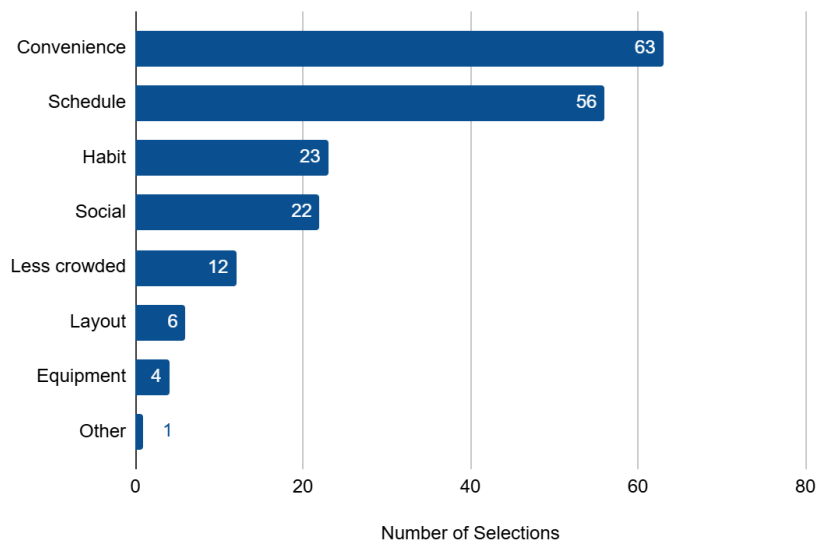


Figure 5. Reasons for Facility Selection

Utilization and Experience Comparisons

When utilization, perceived experience, and equipment access are considered together, a consistent pattern emerges across all conditions. As shown in Figure 1A, average utilization remained low, ranging from 10.5% to 17.2% across facilities. Despite this, Figure 2 shows that perceived crowding remained within a moderate range, with values between 2.4 and 3.0, and increased slightly during higher-demand periods like NorthREC at 5PM. At the same time, Figure 3 indicates that access limitations became more obvious under these same conditions, with the percentage of users experiencing equipment inaccessibility increasing from as low as 13.6% at 11AM to as high as 35.7% at 5PM. These results show that even small increases in usage correspond with noticeable changes in perceived crowding and access to equipment.

These relationships have been summarized in Table 2, which aligns the utilization, perception, and access data within the same conditions. Across all cases, low utilization corresponds with moderate perceived crowding and measurable access limitations, especially during the 5PM block. When considered alongside the uneven machine-level distribution shown in Table 1, these results further show that specific machines or areas experience higher concentrations of use, which shape both perceived crowding and user behavior. As a result, localized conditions within the facility appear to play a more significant role in defining experience than overall usage patterns alone.

Table 2. Comparison of Utilization, Perceived Experience, and Equipment Access by Facility and Time Block

Gym-Time	Average Utilization (%)	Average Crowding	Average Comfort	Average Movement	% No Access
North-11AM	12.4%	2.5	3.5	3.5	13.6%
South-11AM	10.5%	2.7	3.3	3.2	16.0%
North-5PM	17.2%	3	3.2	3.3	28.6%
South-5PM	13.1%	2.4	3.3	3.1	35.7%

Discussion

This study examined how machine utilization patterns relate to perceived user experience within the University of Arizona campus recreation facilities. The central finding is that overall system performance and lived user experience did not align in a simple way. Across observed conditions, machine utilization remained low, yet users still reported moderate perceptions of crowding and noticeable limitations in equipment access. That disconnect is important because recreation facilities are often judged through broad indicators such as capacity, occupancy, or equipment availability. Those measures help describe how a system performs at an aggregate level, but they do not fully explain how the space functions once people begin to interact with it. In this case, the recreation centers were not broadly overloaded, yet recurring friction still shaped the workout experiences of users. The results suggest that performance in fitness environments should be understood not only through how many resources exist, but through how those resources are organized and experienced during actual use. (Thach et al., 2020; Kim et al., 2022).

A major reason for this disconnect appears to be the concentration of demand within specific zones and around specific machine types. Use consistently centered around treadmills and stair-based equipment, while other categories remained comparatively underutilized. This indicates that the issue was not total equipment supply alone, but repeated demand for a narrower set of machine types. User behavior further reinforced that pattern, as rather than redistributing evenly across the facility when access was limited, many users chose to wait, adjust locally, or remain near their preferred zone. This created frequent pockets of congestion within a system that otherwise retained machine availability and capacity. Prior research on clustering behavior suggests that people often gravitate toward familiar, socially active, or environmentally favorable areas rather than optimizing across facilities for total availability (Shin et al., 2013; Cao et al., 2025). The present findings extend that idea into the recreation setting by showing how routine preferences can shape operational outcomes. Layout, visibility, circulation, and adjacency may therefore matter as much as raw equipment counts when evaluating user access.

The perception data further supports this interpretation. Crowd ratings remained moderate, while comfort and ease of movement were generally positive. Those responses are not contradictory when viewed through the lens of spatial experience. A user may experience one cardio zone as crowded, encounter waits for a treadmill, or move through a congested circulation point, while still feeling physically comfortable within the larger scheme of the facility. They may benefit from adequate airflow, acceptable and more comfortable temperatures, daylighting, or open movement in surrounding areas even while a specific zone experiences higher demand. Indoor Environmental Quality research has long emphasized that comfort emerges through the interaction between multiple environmental and spatial factors rather than a single condition (Fissore et al., 2023; Huang et al., 2022). This study suggests that those relationships should be expanded further to include behavioral density and demand concentration. Users are not only responding to temperature, lighting, or air circulation. They are also responding to where other people are, where access is limited, and where additional effort may be required to continue their planned activities.

These findings are especially relevant when considering sustainable design frameworks such as LEED, particularly given that SouthREC operates as a LEED Platinum facility while NorthREC does not

carry the same designation. LEED certification places strong emphasis on indoor environmental quality through strategies like ventilation effectiveness, daylight access, thermal regulation, and material health. In recreation settings, those features are especially meaningful because exercise increases sensitivity through heat, airflow, and general comfort. Within that context, SouthREC's certification likely reflects environmental conditions that can support a more comfortable user experience. The generally positive ratings for comfort and ease of movement observed throughout the study may partially reflect the value of these design strategies, especially in a high-intensity environment where discomfort can be felt more immediately. At the same time, the comparison between SouthREC and NorthREC suggests that environmental certification alone does not fully determine how a recreation center is experienced in practice. User reporting of crowding and equipment inaccessibility indicates that while LEED-oriented strategies may improve environmental quality, they do not automatically resolve operational issues created through layout, circulation patterns, or concentrated demand around machines. A facility may perform strongly within environmental metrics while still producing consistent user friction if preferred equipment is clustered in one area or if movement through the space becomes constrained.

This distinction points toward a broader understanding of sustainability in fitness environments. Environmental performance remains important, but user functionality should also be considered as part of long-term building success. If available equipment remains underused while demand repeatedly concentrates around a few resources, the issue is no longer just comfort but spatial efficiency and how effectively the facility can support engagement. Prior research has shown that occupant satisfaction often depends on how conditions are distributed spatially rather than whether technical thresholds are simply met (Lawal et al., 2025). The present findings extend that argument by suggesting that sustainable recreation design should evaluate both energy and environmental metrics, along with how well a facility aligns its physical layout with actual patterns of use. One of the challenges with recreation center design is that these spatial conditions become relatively fixed once building construction is complete.. Equipment needs to be organized not only by machine type, but also around physical circulation paths, occupancy density, structural limits, safety clearances, and how zones are dispersed within the facility. As a result, patterns of congestion can become embedded into the building's layout itself if high-demand equipment is concentrated within a limited area. This is particularly important in facilities that emphasize sustainable or LEED-oriented design, where environmental conditions such as daylighting, airflow, and thermal comfort may already vary spatially throughout the building. If favorable conditions are not actually aligned with patterns of demand, facilities may unintentionally create recurring congestion points that are difficult to resolve following construction. Thoughtful equipment placement during the initial design process may therefore represent an important, but overlooked, component of long-term user comfort and functional sustainability.

User decision-making also helps explain why these patterns can persist. as facility selection was primarily driven by convenience and schedule, with relatively little emphasis placed on layout, equipment availability, or environmental conditions. Many users appear to choose the facility that best fits their routine, and choose to manage inefficiencies once inside. This suggests that design may have a stronger influence on retention, satisfaction, and day-to-day experience than on initial attraction. From a behavioral perspective, this aligns with Self-Determination Theory, where sustained participation is shaped by perceived competence, autonomy, and comfort during activity (Teixeira et al., 2012; Geller et

al., 2018). Repeated waiting periods, constrained choices, or persistent congestion may gradually decrease favorable conditions even when occupancy remains stable.

Several limitations should be acknowledged. This study was conducted across two University of Arizona recreation centers, which may limit broader generalizability. Data collection was restricted to two time blocks and cannot capture the full range of daily, weekly, or seasonal variation. Survey responses were self-reported and shaped by individual perception, and the intercept method may introduce sampling bias. Future research could expand across additional facilities, include more frequent or longer observation windows, and integrate quantitative data on environmental metrics such as temperature, airflow, or daylight intensity alongside behavioral data. Despite these limitations, the consistency of patterns across both facilities and time blocks supports the reliability of the findings.

Conclusion

This study examined how machine utilization patterns and perceived user experience varied across two University of Arizona recreation centers, with a broader focus on how system demand, spatial organization, and environmental conditions interact within fitness environments. To address that question, the research combined structured observations of cardiovascular equipment usage with intercept survey responses measuring crowding, comfort, movement, and equipment access. The results showed that overall machine utilization remained low across both recreation centers, yet users still reported moderate perceptions of crowding and noticeable limitations in access. Machine-level analysis further revealed that demand concentrated around specific equipment types, particularly treadmills and stair-based machines, while other equipment remained underutilized. These patterns indicate that user experience is shaped less by total system capacity and more by how demand is distributed throughout the facility. As a result, conditions of localized congestion can emerge even when the facility is operating well below its overall capacity.

More broadly, the study shows the importance of evaluating recreation centers through both performance metrics and lived user experience. While sustainable design strategies such as LEED aim to improve indoor environmental quality and support baseline comfort, they do not fully account for how spaces are used and experienced in practice. Layout and equipment distribution appear to shape user experience more directly because they influence how demand concentrates, how users move through the facility, and where recurring friction develops during usage. For designers, planners, and facility operators, this suggests that improvements may depend less on expansion and more on aligning the existing facilities with actual patterns of use. Understanding these relationships may help inform more effective design and planning strategies for recreation centers that better distribute demand, reduce recurring congestion, and align environmental quality with actual patterns of use.

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